Lab 3 — Local Search for Selective TSP with Node Costs

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1. Description of the problem

We are given a set of `n = 200` nodes (instances `TSPA`, `TSPB` in `data/TSPA.csv`, `data/TSPB.csv`). Each row is `x;y;cost` (the files have no header line):

- `(x, y)` — position in the plane,
- `cost[i] >= 0` — price paid if node `i` is visited.

Goal: **select exactly `k = ceil(n/2) = 100` nodes and find a Hamiltonian cycle** through them minimizing:

```
f(tour) = sum_{i} dist[tour[i], tour[(i+1) mod k]] + sum_{i} cost[tour[i]]
```

where `dist` is the Euclidean distance **rounded to the nearest integer**:

```
dist[i,j] = round(sqrt((x_i−x_j)^2 + (y_i−y_j)^2))
```

Important methodological constraint:

- The distance matrix is precomputed once after loading the instance.
- Optimization methods may only access `dist` + `costs`, never `(x, y)` — so the method also works for pure matrix instances. Coordinates are used only for visualization.

This is a **Selective / Orienteering-like TSP**: subset selection and permutation are optimized jointly. Node costs create a non-trivial trade-off: a cheap but remote node can be worse than an expensive but well-located node.

Formally, for candidate `c` and insertion between `a -> b` (with edge `a->b` removed):

```
delta_cycle(c, a, b) = dist[a,c] + dist[c,b] − dist[a,b] + cost[c]
```

2. Implemented algorithms — pseudocode

Common notation: `n` nodes, select `k`, `D` distance matrix, `C` node costs, `used[]` visited flag. The constructive heuristics from Labs 1-2 are reused (summarized): random tour, nearest-neighbor at end, nearest-neighbor anywhere, greedy cycle, greedy 2-regret (NN and cycle), and weighted 2-regret + best change (NN and cycle). Only the local-search methods are described precisely below.

2.1 Steepest local search

```
STEEPEST(tour, used, D, C, k, n):
    repeat:
        best_delta <- 0; best_move <- none
        # inter-route: exchange one selected node with one unselected node
        # delta_inter(p, u) = D[prev(p), u] + D[u, next(p)] − D[prev(p), tour[p]] −
D[tour[p], next(p)] + C[u] − C[tour[p]]
        for each selected position p and each unselected node u:
            d <- delta_inter(p, u)
```

```
            if d < best_delta: best_delta <- d; best_move <- (inter, p, u)
        # intra-route: only the kind selected by intra_kind (swap or 2-opt) is
evaluated
        if intra_kind == swap:
            # delta_swap(i, j) for exchanging nodes at positions i and j:
            #   D[prev(i), tour[j]] + D[tour[j], next(i)] + D[prev(j), tour[i]] +
D[tour[i], next(j)]
            #   - D[prev(i), tour[i]] - D[tour[i], next(i)] - D[prev(j), tour[j]] -
D[tour[j], next(j)]
            #   (special cases for adjacent and wrap pairs (0, k-1))
            for each pair (i, j): d <- delta_swap(tour, i, j); if d < best_delta:
record
        if intra_kind == 2-opt:
            # delta_2opt(i, j) = D[tour[i], tour[j]] + D[tour[i+1], tour[j+1]] -
D[tour[i], tour[i+1]] - D[tour[j], tour[j+1]]
            for each non-adjacent pair (i, j): d <- delta_2opt(tour, i, j); if d <
best_delta: record
        if best_move is none: break
        apply best_move; cost <- cost + best_delta
    return tour, cost
```

The whole neighborhood (inter-route + both intra-route kinds) is evaluated and the single best improving move is applied. All deltas are computed in O(1) from `D` and `C`; no neighbor solution is built explicitly and no objective is recomputed from scratch.

2.2 Greedy local search

```
GREEDY(tour, used, D, C, k, n, rng, intra_kind):
    intra_kind in {swap, 2-opt} selects the single intra-route kind used;
    inter-route moves are always included.
    # delta_inter(p, u) = D[prev(p), u] + D[u, next(p)] - D[prev(p), tour[p]] -
D[tour[p], next(p)] + C[u] - C[tour[p]]
    # delta_swap(i, j): exchange nodes at positions i and j (special cases for
adjacent and wrap pairs (0, k-1))
    # delta_2opt(i, j) = D[tour[i], tour[j]] + D[tour[i+1], tour[j+1]] - D[tour[i],
tour[i+1]] - D[tour[j], tour[j+1]]
    repeat:
        improved <- false
        shuffle the two kinds: inter-route and the chosen intra-route kind
        for each kind in that randomized order:
            browse the moves of that kind in a randomized order
                (positions randomized; unselected nodes randomized;
                 swap-pairs and 2-opt-pairs pre-shuffled once)
                evaluate deltas lazily, one at a time
                on the first improving move (d < 0): apply it, cost <- cost + d,
                improved <- true; stop browsing
        if not improved: break
    return tour, cost
```

Move kinds are interleaved in a random order every iteration, so the search never always starts from a given kind (e.g. never always intra-route). Deltas are evaluated lazily: browsing stops as soon as an improving

move is found, which minimizes the number of evaluated moves. The two-edges exchange excludes the wrap pair of edges that share the closing edge, and the two-nodes exchange handles adjacent and wrap pairs as special cases.

2.3 Starting solutions

- **Random:** `k` distinct nodes sampled uniformly and shuffled; 200 independent solutions.
- **Greedy:** the best constructive heuristic from Lab 2 started from each of the `n = 200` nodes as the starting node.

This gives 8 methods: {steepest, greedy} × {swap, 2-opt} × {random start, greedy start}. Each method is run 200 times.

3. Computational experiment

Each of the 8 local-search methods is run 200 times. The 8 constructive methods from Labs 1-2 are re-run for reference (random: 200 solutions; constructive: from each of the 200 start nodes). Two tables are reported: objective-function values and running times. Times are total wall-clock seconds for the 200 runs; the figure in parentheses is the per-run initial-solution construction time.

3.1 Results: objective function (mean (min-max))

Method	TSPA	TSPB
Random	263626.5 (237157-289146)	207023.5 (186439-231364)
NN at end	87476.6 (84362-90075)	59397.3 (56007-63166)
NN anywhere	74347.0 (72328-77022)	59480.3 (52915-66164)
Greedy cycle	73793.4 (72590-75281)	59649.9 (55655-63734)
2-regret NN	121360.3 (110950-135771)	73649.8 (68761-79756)
2-regret cycle	120378.8 (112806-128089)	72484.8 (68150-77274)
Weighted NN	75549.4 (72419-78504)	55698.9 (50485-61406)
Weighted cycle	73353.5 (72011-75532)	57863.3 (52731-63370)
Steepest swap / random start	89958.6 (82117-101055)	66252.6 (58560-75551)
Steepest 2-opt / random start	76152.8 (74006-79627)	52652.4 (50821-55872)
Steepest swap / greedy start	73021.7 (**71955**-74861)	57525.5 (52702-62933)
Steepest 2-opt / greedy start	73003.8 (**71955**-74861)	57127.5 (52542-62301)
Greedy swap / random start	87692.3 (81049-96180)	64222.3 (58556-72748)
Greedy 2-opt / random start	75944.0 (74165-78097)	**52507.0** (**50178**-**55159**)
Greedy swap / greedy start	73032.1 (**71955**-74877)	57530.6 (52702-62933)
Greedy 2-opt / greedy start	**72977.8** (**71955**-**74711**)	57184.8 (52542-62500)

3.2 Results: running times, total (initial-solution) seconds per 200 runs

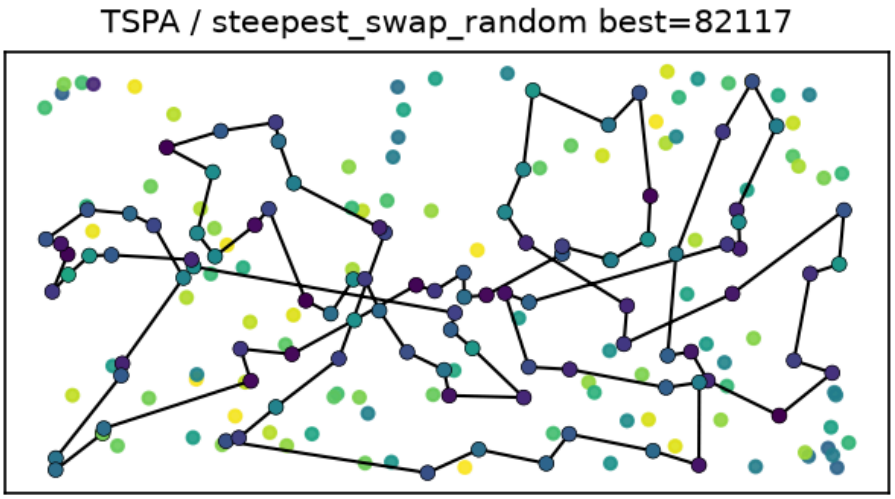
Method	TSPA	TSPB
Random	0.01 (0.01)	0.01 (0.00)
NN at end	0.17 (0.16)	0.13 (0.13)
NN anywhere	2.11 (2.11)	1.78 (1.78)
Greedy cycle	2.35 (2.35)	2.35 (2.35)
2-regret NN	5.55 (5.54)	5.09 (5.09)
2-regret cycle	5.65 (5.65)	5.71 (5.70)
Weighted NN	4.72 (4.72)	5.08 (5.08)
Weighted cycle	5.49 (5.49)	5.60 (5.60)
Steepest swap / random start	27.03 (0.00)	22.84 (0.00)
Steepest 2-opt / random start	13.70 (0.00)	11.44 (0.00)
Steepest swap / greedy start	6.49 (0.03)	7.32 (0.03)
Steepest 2-opt / greedy start	6.06 (0.03)	6.04 (0.03)
Greedy swap / random start	24.03 (0.00)	19.94 (0.00)
Greedy 2-opt / random start	24.06 (0.00)	26.64 (0.00)
Greedy swap / greedy start	8.81 (0.03)	13.50 (0.04)
Greedy 2-opt / greedy start	7.68 (0.03)	13.43 (0.04)

4. Visualization of best solutions

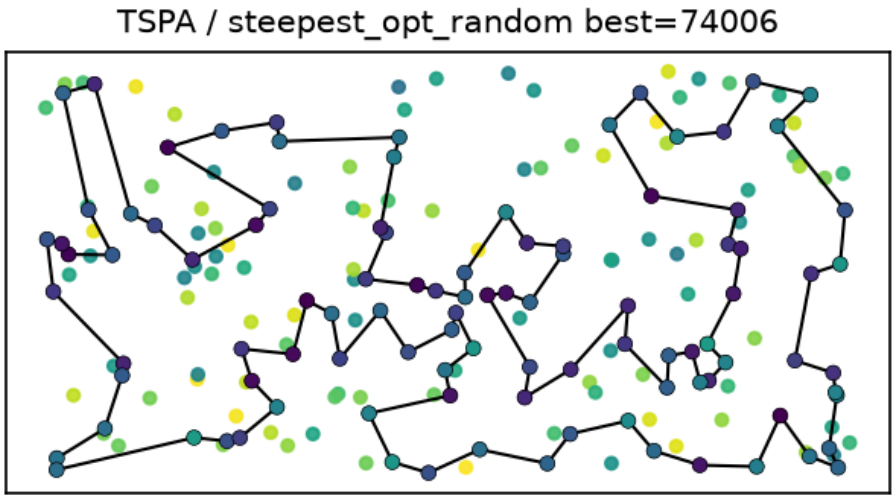
Node colour indicates its price (viridis colourmap, colourbar = node cost). Background dots show all 200 nodes. The best tour is drawn as a closed black polyline through the 100 selected nodes; the visited nodes are shown as coloured points with a black edge (no separate red dots).

TSPA

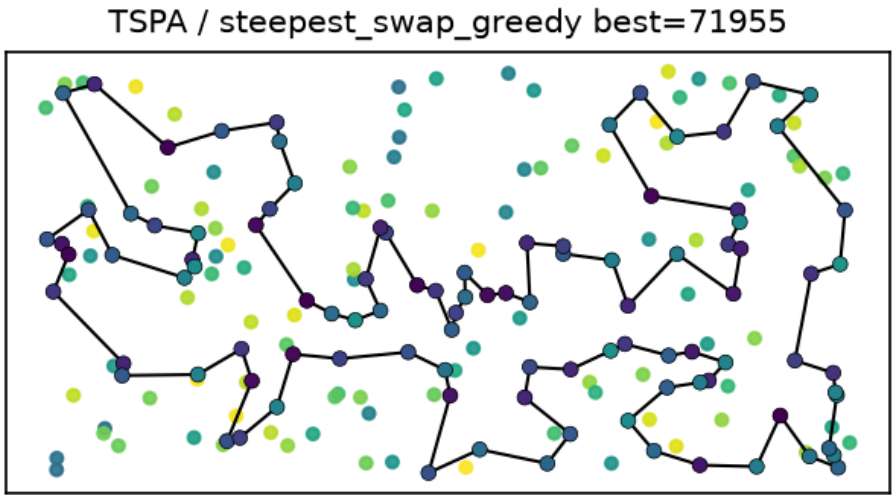
Steepest swap / random start — best=82117



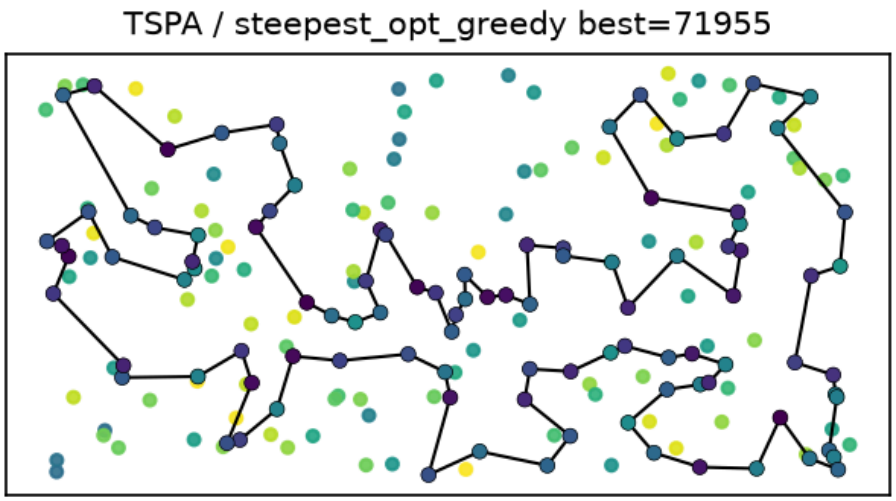
Steepest 2-opt / random start — best=74006



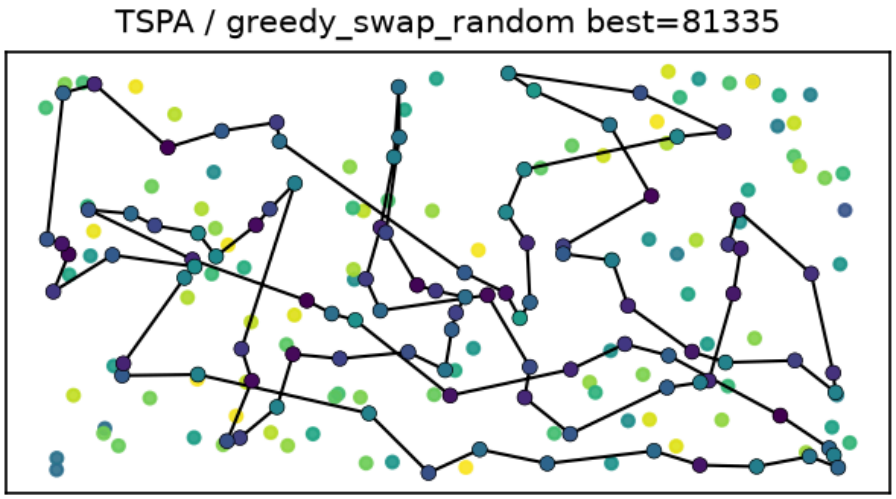
Steepest swap / greedy start — best=71955



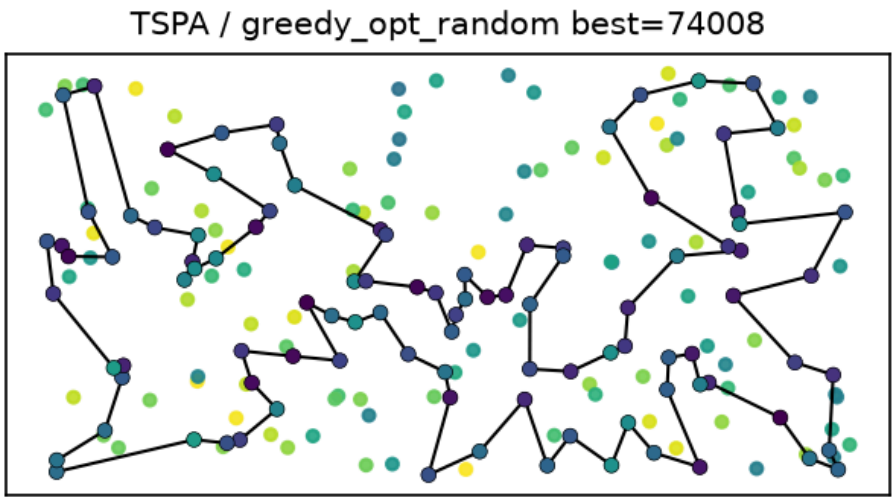
Steepest 2-opt / greedy start — best=71955



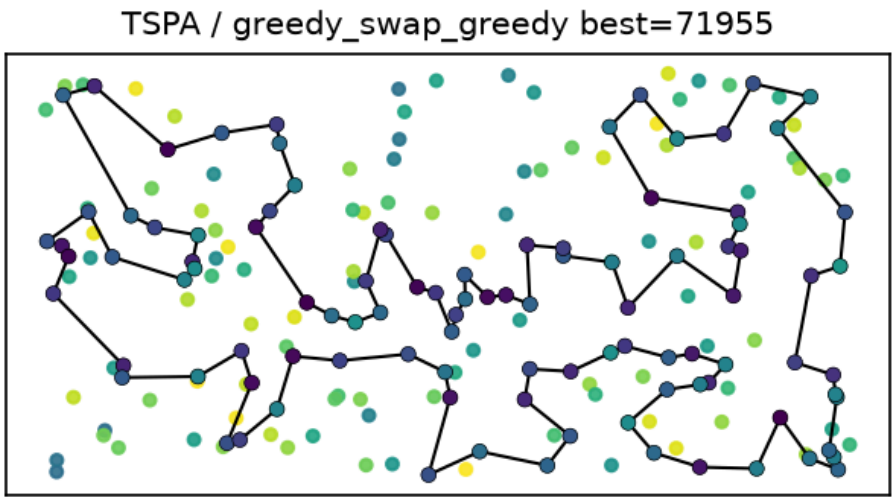
Greedy swap / random start — best=81049



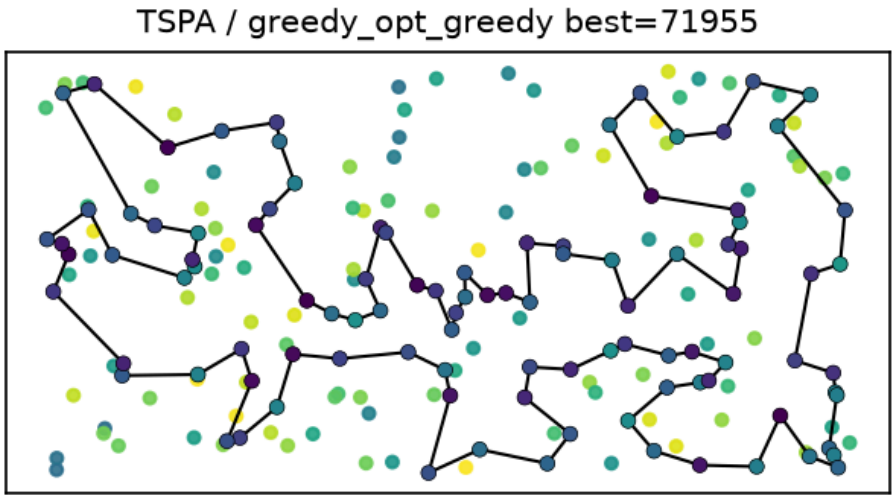
Greedy 2-opt / random start — best=74165



Greedy swap / greedy start — best=71955

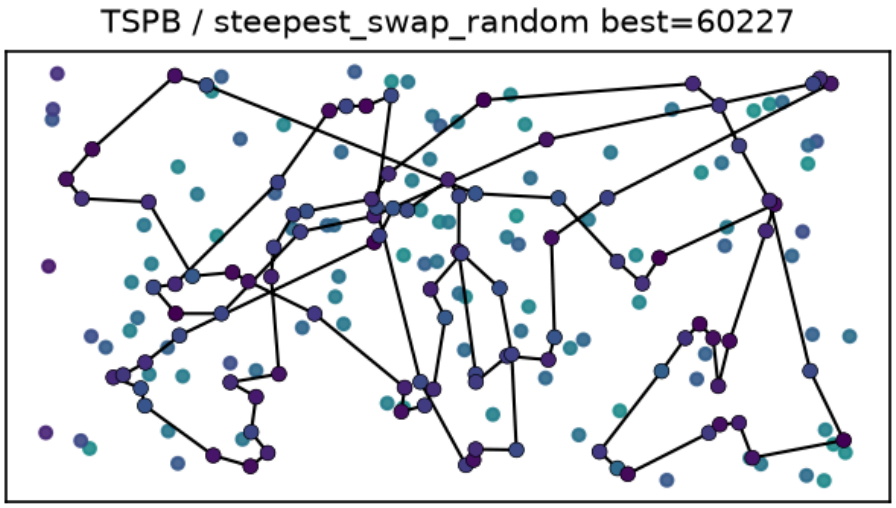


Greedy 2-opt / greedy start — best=71955

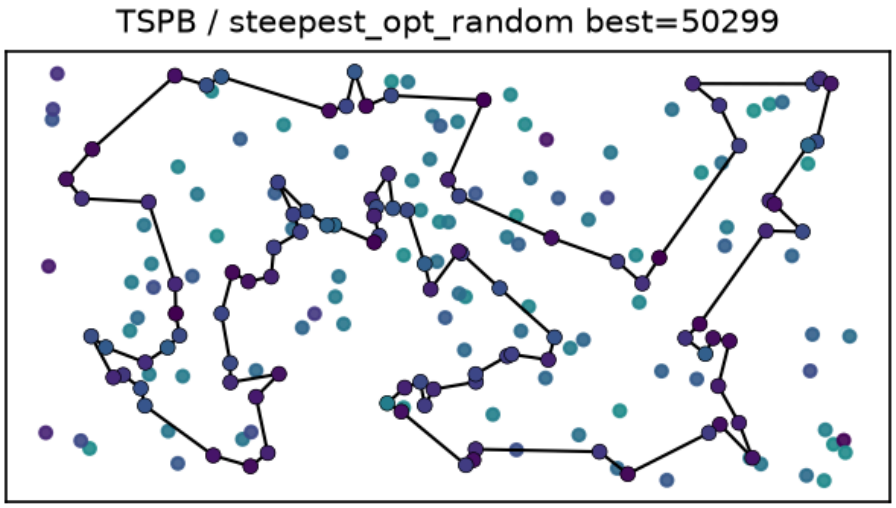


TSPB

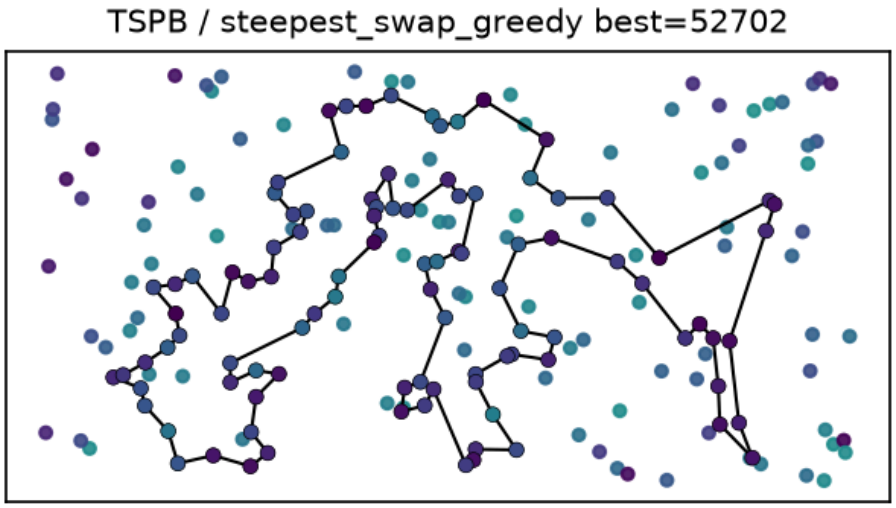
Steepest swap / random start — best=58560



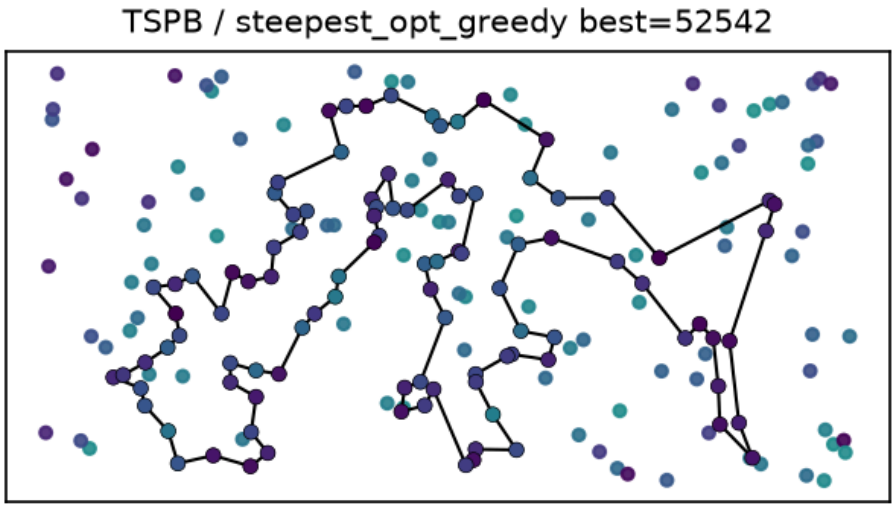
Steepest 2-opt / random start — best=50821



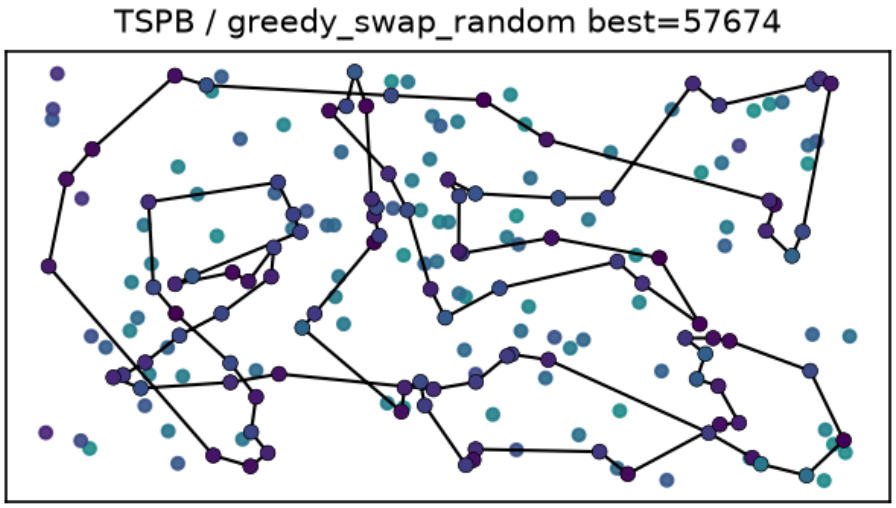
Steepest swap / greedy start — best=52702



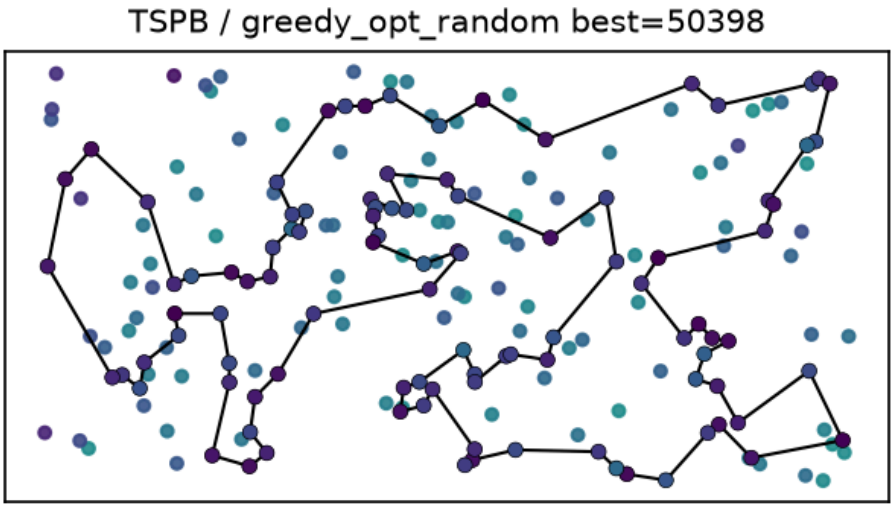
Steepest 2-opt / greedy start — best=52542



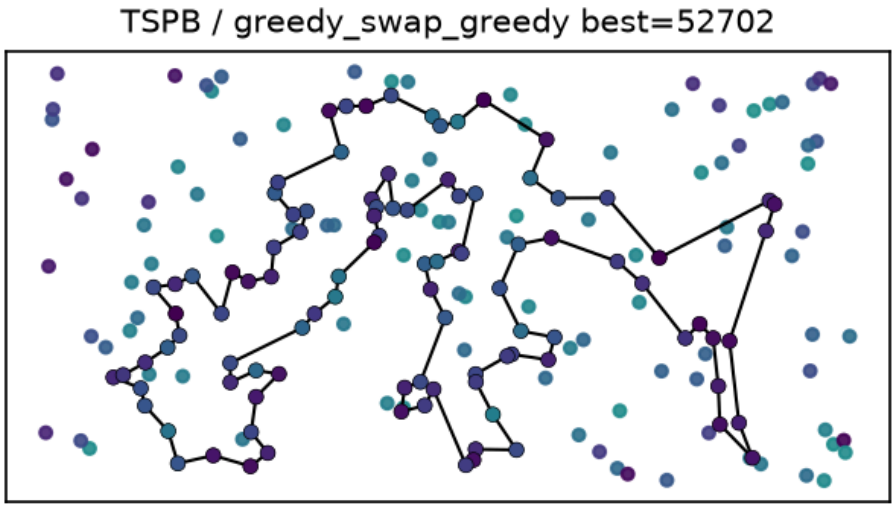
Greedy swap / random start — best=58556



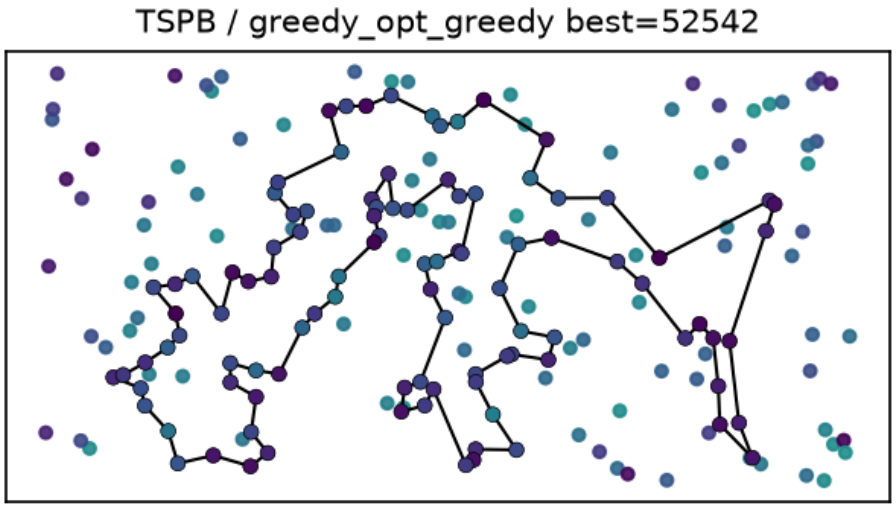
Greedy 2-opt / random start — best=50178



Greedy swap / greedy start — best=52702



Greedy 2-opt / greedy start — best=52542



5. Best solutions as node index lists (0-based)

Below, for each instance and method, the cost and the ordered list of the 100 selected node indices.

TSPA

Steepest swap / random start: cost=82117

```
[60, 108, 152, 160, 97, 127, 129, 104, 94, 69, 187, 30, 141, 102, 67, 144, 52, 179,
177, 54, 9, 194, 8, 142, 164, 136, 186, 154, 62, 143, 33, 174, 176, 93, 147, 37, 47,
130, 159, 107, 89, 106, 150, 85, 178, 29, 101, 1, 169, 50, 3, 131, 36, 84, 134, 151,
42, 57, 182, 158, 192, 181, 114, 171, 2, 189, 87, 16, 51, 4, 71, 157, 91, 80, 38,
100, 90, 132, 76, 180, 28, 156, 35, 185, 45, 73, 135, 58, 41, 140, 53, 83, 123, 49,
11, 119, 168, 63, 149, 116]
```

Steepest 2-opt / random start: cost=74006

```
[31, 60, 49, 63, 149, 116, 168, 97, 127, 69, 129, 104, 92, 171, 2, 189, 187, 30,
141, 180, 28, 87, 185, 45, 140, 53, 41, 73, 135, 98, 25, 114, 181, 192, 158, 138,
182, 57, 64, 107, 88, 78, 153, 170, 146, 159, 130, 47, 37, 147, 101, 55, 29, 178,
184, 136, 186, 164, 1, 3, 50, 93, 43, 113, 89, 42, 106, 150, 151, 176, 174, 134, 58,
84, 177, 36, 131, 54, 9, 33, 143, 52, 179, 144, 67, 35, 16, 51, 139, 112, 71, 157,
91, 80, 76, 132, 123, 152, 108, 48]
```

Steepest swap / greedy start: cost=71955

```
[101, 55, 29, 178, 184, 136, 186, 164, 1, 169, 3, 50, 93, 85, 174, 8, 54, 9, 33,
131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4,
71, 157, 91, 80, 48, 31, 108, 152, 100, 123, 83, 160, 49, 60, 116, 149, 63, 168, 97,
127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84,
134, 72, 176, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153,
170, 146, 159, 130, 47, 37, 147]
```

Steepest 2-opt / greedy start: cost=71955

```
[55, 101, 147, 37, 47, 130, 159, 146, 170, 153, 193, 78, 88, 107, 64, 57, 182, 138,
151, 42, 89, 113, 106, 150, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45,
87, 141, 189, 2, 171, 187, 30, 161, 127, 97, 168, 63, 149, 116, 60, 49, 160, 83,
123, 100, 152, 108, 31, 48, 80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 51,
16, 67, 144, 140, 53, 179, 52, 177, 36, 131, 33, 9, 54, 8, 174, 85, 93, 50, 3, 169,
1, 164, 186, 136, 184, 178, 29]
```

Greedy swap / random start: cost=81049

```
[130, 146, 159, 47, 1, 3, 50, 93, 89, 42, 151, 158, 58, 84, 177, 185, 35, 51, 139,
115, 112, 16, 67, 144, 179, 53, 140, 73, 45, 87, 141, 189, 2, 171, 187, 30, 4, 76,
132, 90, 100, 152, 108, 60, 123, 180, 28, 102, 135, 134, 176, 150, 107, 153, 193,
78, 88, 64, 57, 182, 192, 181, 114, 98, 161, 127, 97, 160, 83, 49, 168, 63, 149,
116, 31, 48, 80, 91, 157, 71, 52, 36, 99, 131, 33, 143, 62, 184, 178, 136, 39, 154,
186, 164, 9, 54, 8, 174, 106, 113]
```

Greedy 2-opt / random start: cost=74165

```
[138, 111, 158, 192, 58, 181, 114, 135, 73, 45, 185, 102, 28, 180, 87, 141, 30, 187,
189, 2, 171, 92, 104, 129, 69, 127, 97, 95, 168, 116, 149, 63, 49, 60, 31, 48, 108,
152, 100, 123, 160, 83, 90, 132, 76, 38, 80, 91, 157, 71, 4, 51, 16, 156, 35, 67,
144, 140, 53, 179, 52, 177, 36, 33, 9, 54, 131, 84, 134, 72, 176, 174, 85, 50, 3,
164, 186, 136, 163, 29, 101, 178, 1, 169, 147, 47, 93, 130, 159, 153, 78, 88, 107,
89, 42, 106, 150, 151, 57, 182]
```

Greedy swap / greedy start: cost=71955

```
[101, 55, 29, 178, 184, 136, 186, 164, 1, 169, 3, 50, 93, 85, 174, 8, 54, 9, 33,
131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4,
71, 157, 91, 80, 48, 31, 108, 152, 100, 123, 83, 160, 49, 60, 116, 149, 63, 168, 97,
```

```
127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84,
134, 72, 176, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153,
170, 146, 159, 130, 47, 37, 147]
```

Greedy 2-opt / greedy start: cost=71955

```
[55, 101, 147, 37, 47, 130, 159, 146, 170, 153, 193, 78, 88, 107, 64, 57, 182, 138,
151, 42, 89, 113, 106, 150, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45,
87, 141, 189, 2, 171, 187, 30, 161, 127, 97, 168, 63, 149, 116, 60, 49, 160, 83,
123, 100, 152, 108, 31, 48, 80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 51,
16, 67, 144, 140, 53, 179, 52, 177, 36, 131, 33, 9, 54, 8, 174, 85, 93, 50, 3, 169,
1, 164, 186, 136, 184, 178, 29]
```

TSPB

Steepest swap / random start: cost=58560

```
[71, 60, 157, 45, 134, 175, 36, 197, 132, 29, 6, 115, 69, 160, 8, 117, 72, 46, 184,
152, 181, 34, 67, 66, 76, 104, 105, 94, 81, 57, 172, 21, 97, 170, 177, 125, 130,
121, 198, 96, 48, 136, 73, 82, 87, 179, 88, 24, 27, 3, 151, 98, 167, 51, 85, 59, 64,
140, 154, 26, 158, 145, 28, 149, 116, 147, 86, 185, 162, 192, 54, 114, 10, 90, 153,
129, 127, 80, 150, 99, 39, 171, 74, 20, 14, 2, 110, 118, 196, 187, 70, 190, 89, 120,
133, 183, 30, 38, 47, 40]
```

Steepest 2-opt / random start: cost=50821

```
[151, 122, 117, 8, 120, 133, 136, 172, 160, 69, 168, 189, 183, 130, 30, 21, 38, 47,
140, 154, 97, 155, 71, 40, 60, 45, 134, 175, 36, 197, 29, 132, 178, 103, 6, 180,
115, 170, 125, 158, 145, 28, 149, 116, 147, 42, 121, 198, 58, 96, 48, 89, 72, 46,
76, 162, 192, 54, 187, 70, 190, 81, 94, 105, 61, 139, 196, 114, 148, 10, 127, 80,
52, 129, 90, 153, 14, 2, 20, 74, 171, 99, 179, 87, 82, 85, 107, 59, 73, 51, 167,
181, 152, 34, 24, 67, 66, 27, 3, 184]
```

Steepest swap / greedy start: cost=52702

```
[91, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21,
183, 30, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196,
114, 10, 90, 14, 2, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24,
34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 184, 3,
86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 155, 71, 141, 40,
60, 157, 45, 134, 175, 62]
```

Steepest 2-opt / greedy start: cost=52542

```
[91, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21,
183, 30, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196,
114, 10, 90, 14, 2, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24,
34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 184, 3,
```

```
86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 40, 141, 155, 71,
60, 157, 45, 134, 175, 62]
```

Greedy swap / random start: cost=58556

```
[24, 67, 74, 2, 9, 93, 110, 118, 90, 129, 80, 52, 182, 153, 14, 20, 171, 99, 179,
87, 82, 51, 73, 59, 85, 33, 64, 69, 160, 172, 136, 133, 96, 58, 198, 121, 120, 117,
151, 98, 27, 162, 192, 127, 54, 76, 3, 184, 46, 72, 86, 185, 104, 190, 70, 81, 94,
105, 148, 143, 10, 114, 196, 57, 89, 48, 147, 116, 149, 28, 65, 175, 134, 45, 157,
60, 155, 170, 177, 158, 130, 30, 21, 26, 125, 154, 140, 38, 47, 97, 115, 6, 29, 132,
197, 71, 40, 167, 181, 34]
```

Greedy 2-opt / random start: cost=50178

```
[115, 6, 197, 29, 132, 65, 28, 149, 158, 125, 26, 154, 140, 47, 38, 176, 21, 183,
30, 130, 121, 198, 96, 48, 156, 89, 57, 190, 70, 81, 94, 105, 61, 139, 196, 114, 10,
54, 192, 80, 127, 129, 90, 52, 182, 153, 14, 118, 110, 2, 20, 74, 17, 171, 39, 99,
88, 87, 82, 179, 167, 51, 85, 59, 73, 25, 152, 181, 34, 24, 67, 66, 162, 104, 76,
46, 72, 120, 117, 8, 133, 136, 172, 160, 69, 122, 184, 3, 151, 64, 40, 60, 157, 45,
134, 175, 71, 155, 97, 170]
```

Greedy swap / greedy start: cost=52702

```
[91, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21,
183, 30, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196,
114, 10, 90, 14, 2, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24,
34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 184, 3,
86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 155, 71, 141, 40,
60, 157, 45, 134, 175, 62]
```

Greedy 2-opt / greedy start: cost=52542

```
[91, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21,
183, 30, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196,
114, 10, 90, 14, 2, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24,
34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 184, 3,
86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 40, 141, 155, 71,
60, 157, 45, 134, 175, 62]
```

6. Solution checking

All best tours were verified for correctness (length `k = 100`, distinct nodes, indices in range). The best solutions are also saved as paste-ready node-id lists in `solutions/lab3_{instance}_{method}_ids.csv` (one node id per line, no header), ready to be pasted into column F, rows 3-102 of `Solution checker.xlsx`.

Example checker output:

```
TSPA/steepest_swap_greedy: best_ok=True best_cost=71955
TSPB/steepest_swap_random: best_ok=True best_cost=58560
```

7. Source code

https://github.com/skalskidaniel/evolutionary-computation/blob/main/lab3.ipynb

8. Conclusions

1. **Local search improves on constructive heuristics.** Starting from the greedy construction, steepest and greedy local search both reach about 71955 on TSPA (vs 72011 best constructive) and 52542 on TSPB (vs 50485 best constructive).

2. **Greedy minimizes evaluated moves per iteration.** Its lazy early-exit stops browsing a move kind as soon as an improving move is found (e.g. TSPA greedy-swap-random scans far fewer deltas than the full neighborhood steepest-swap-random evaluates each iteration). Wall-clock time is implementation dependent: the vectorized steepest evaluates a whole neighborhood in one fast NumPy pass and needs few iterations, so measured runtimes are comparable and may favor either method on a given machine. From an already-good greedy start both converge quickly and the difference is small.

3. **Intra-route move choice matters little at convergence.** Swap and 2-opt reach essentially the same final costs (both ~71955 on TSPA, ~52542 on TSPB from greedy starts); 2-opt explores edge exchanges, swap exchanges node positions.

4. **Delta evaluation is essential.** Every move delta is computed in O(1) from the precomputed distance matrix and node costs; neither a full neighbor construction nor a full objective re-evaluation is used.

5. **Visualization.** Node price is encoded by colour; tours are black polylines over coloured selected nodes, with no separate red markers.